Impact of Dryocosmus kuriphilus on Castanea sativa

Physiological and Morphological Leaf Trait Analysis

Five Provenances — Three Countries

University Forest of Taxiarchis, Halkidiki, Greece (760 m a.s.l.)

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R Analysis Walkthrough — Portfolio Documentation

1. Description and Disclaimer

1.1 Study Overview

This script accompanies a study examining the effects of the Asian chestnut gall wasp (Dryocosmus kuriphilus) on European chestnut (Castanea sativa) trees. The gall wasp is an invasive pest that has caused significant ecological and economic damage to chestnut populations across Europe, stimulating interest in provenance-level differences in susceptibility and physiological response.

The analysis focuses on two interrelated datasets collected in a common-garden experiment at the University Forest of Taxiarchis (Halkidiki, Greece, 760 m a.s.l.):

- Branch-level infestation data — used to calculate the Damage and Crown Index (DCI), plus counts of reactivated dormant buds (RDB), dead shoots, and galls.
- Leaf-level physiological and morphological trait data — including chlorophyll content (CCI), photosystem II efficiency (Fv/Fm), performance index (PI), stomatal conductance (Gs), transpiration (Trm), light-saturated photosynthesis (A_sat), leaf mass per area (LMA), leaf dry matter content (LDMC), and leaf weight/area measurements.

Five chestnut provenances from three countries are compared:

- Spain: Coruna, Malaga
- Greece: Hortiatis
- Italy: Pellice, Sicily

1.2 Disclaimer

> *IMPORTANT: This script runs entirely on synthetic data. Real field measurements remain the exclusive property of the University of the Aegean / ELGO Demeter / Aristotle University of Thessaloniki. The code has been rewritten from scratch to avoid copyright issues. The synthetic dataset is designed to mimic the structure and statistical properties of the original data for demonstration purposes only.*

The DCI formula follows the method of Gehring et al. (2018) as published in the Journal of Pest Science. All statistical analyses are implemented in R. Figures are exported as PNG files suitable for inclusion in reports and publications.

2. Methodology: Step-by-Step Code Walkthrough

Step 0 — Package Setup

The script begins by declaring all required R packages in a character vector. A for-loop checks each package using requireNamespace() and installs any that are missing, avoiding errors when the script is run on a fresh machine. All libraries are then loaded silently to suppress startup messages that would clutter console output.

Packages used and their roles:

- ggplot2 — core plotting engine for all figures
- dplyr, tidyr — data manipulation and reshaping
- car — Levene's test for homogeneity of variances
- gplots — supplementary plotting utilities
- FactoMineR, factoextra — PCA computation and visualization
- GGally — correlation matrix heatmap (ggcorr)
- lme4, lmerTest — linear mixed-effects models
- patchwork — side-by-side composition of ggplot figures

```
pkgs <- c("ggplot2", "dplyr", "tidyr", "car", "gplots",
          "FactoMineR", "factoextra", "GGally", "lme4", "lmerTest", "patchwork")
for (p in pkgs) {
  if (!requireNamespace(p, quietly = TRUE)) install.packages(p)
}
```

A shared ggplot2 theme (theme_study) and two named colour palettes (prov_pal for the five provenances, country_pal for the three countries) are defined at the top level so all figures have a consistent visual language throughout the script.

Step 1 — Load Data

Data are loaded by sourcing a companion script generate_synthetic_data.R, which creates two data frames in the global environment:

- dci_synth — branch-level data containing the DCI index and its components (Sd, Bdor, Gons), plus RDB, dead shoots, and gall counts per tree.
- phys_synth — leaf-level physiological and morphological measurements, two leaves per tree.

A clean complete-case subset (phys) is immediately derived by keeping only rows where all twelve physiological variables are non-missing. This subset is used in the mixed-effects models in Step 9. Summary counts are printed to confirm the number of trees and leaf records loaded successfully.

```
source("generate_synthetic_data.R")
dci        <- dci_synth
physiologia <- phys_synth
phys <- physiologia[complete.cases(physiologia[, c("PI","Fv.Fm","CCI",
                     "Gs","Trm","A_sat","F_W","T_W","D_W","L_Area","LDMC","LMA")]), ]
```

Step 2 — DCI Infestation Index

The Damage and Crown Index (DCI) quantifies infestation severity at the tree level. Following Gehring et al. (2018), it is calculated as a weighted linear combination of three branch-level variables:

- Sd — proportion of shoots with dried galls
- Bdor — proportion of buds with dormant galls
- Gons — proportion of buds with active (gonocalyx) galls

```
DCI = (Sd * 0.479 + Bdor * 0.525 + Gons * 0.120) * 100
```

Descriptive statistics (n, mean, SD) are printed by provenance. Two ANOVAs are run — one with Variety and one with Country as the grouping factor. For each, assumption checks include Levene's test for variance homogeneity and a Shapiro-Wilk test on model residuals. A Kruskal-Wallis test is also run as a nonparametric alternative because the DCI distribution may be non-normal.

Figure 1 is produced as a violin + boxplot combination, ordered by median DCI. A white dot marks the mean. The plot is saved to fig1_DCI_provenance.png.

Step 3 — RDB, Dead Shoots, and Gall Counts

Three additional branch-level damage metrics are analysed:

- RDB (Reactivated Dormant Buds) — buds that re-sprouted after gall damage
- Dead_Sh (Dead Shoots) — proportion of shoots that died
- Galls — raw gall count

A reusable helper function run_anova_kw() is defined to avoid code repetition. For a given variable and grouping factor it runs: an ANOVA, Levene's test, Shapiro-Wilk on residuals, and a Kruskal-Wallis test. Results are printed in a single formatted line per variable. The function is called six times in total (three variables x two grouping factors). Figure 2 shows RDB distributions by provenance as a boxplot.

Step 4 — Chlorophyll Content Index (CCI)

CCI is measured with a portable chlorophyll meter and serves as a proxy for leaf photosynthetic capacity. An ANOVA is run with Variety as the factor; if it is significant ($p < 0.05$), Tukey's HSD post-hoc test is applied and only significant pairwise comparisons are printed. A second ANOVA tests the Country factor. Figure 3 combines two panels side by side using patchwork: CCI by provenance (violin + boxplot) and CCI by country.

Step 5 — Photosynthesis (A_sat)

Light-saturated net photosynthesis (A_sat, micromol CO2 m-2 s-1) is analysed after removing NA rows. Three tests are compared: a standard one-way ANOVA, Welch's ANOVA (oneway.test, which does not assume equal variances), and a Kruskal-Wallis test. If the ANOVA is significant, Tukey's HSD pairs are reported. This triple-testing approach is good practice when the data may violate ANOVA assumptions.

Step 6 — Other Leaf Traits

Seven additional leaf traits are tested systematically using the same run_anova_kw() helper from Step 3:

- Fv/Fm — maximum quantum yield of photosystem II (fluorescence)
- LMA — leaf mass per area (g m-2)
- LDMC — leaf dry matter content (mg g-1)
- F_W — fresh leaf weight (g)
- D_W — dry leaf weight (g)
- T_W — saturated (turgid) leaf weight (g)
- L_Area — leaf area (cm2)

Each variable is subset to complete cases before testing to avoid issues caused by sporadic NAs. Results for all seven variables are printed consecutively to the console.

Step 7 — Spearman Correlations: DCI & RDB with Leaf Traits

Spearman rank correlations (rho) are calculated between each of the nine physiological variables and both DCI and RDB. Spearman is chosen over Pearson because DCI and several leaf traits are not normally distributed. For each variable, two cor.test() calls are made and results are printed in a formatted table showing rho and p-value for both infestation indices. Pairs significant at $p < 0.05$ are flagged. A final correlation between RDB and DCI directly is also reported, expected to be strongly positive (rho ~ 0.75) as both measure infestation-related damage.

Step 8 — Principal Component Analysis (PCA)

Before running PCA the leaf data are aggregated to tree level by taking the mean of the two leaf measurements per tree (using dplyr group_by + summarise). This avoids pseudoreplication — using both leaves from the same tree as independent observations in an ordination would artificially inflate sample size and distort the ordination axes.

Two PCAs are run using FactoMineR::PCA():

1. Full PCA — nine traits: PI, Fv/Fm, CCI, Gs, Trm, A_sat, LMA (scaled x1000), LDMC, and DCI. The LMA scaling is purely cosmetic to make the arrow length visually comparable to other traits in the biplot. Variance explained by the first three principal components is printed. Saved as fig4_PCA_full.png.
2. Reduced PCA — six traits: PI, CCI, A_sat, LMA, LDMC, DCI. This subset highlights the ecologically most interpretable traits and their relationship to infestation. Saved as fig5_PCA_reduced.png.

Both biplots use fviz_pca_biplot() from factoextra. Points are coloured by provenance using prov_pal; arrow opacity is scaled by the cos2 value (quality of representation on the displayed axes), so poorly represented variables are shown more transparently.

```
dat_pca <- physiologia %>%
  group_by(ID, Variety) %>%
  summarise(PI = mean(PI, na.rm=TRUE), CCI = mean(CCI, na.rm=TRUE), ...)
pca1 <- PCA(dat_pca[, 3:11], graph = FALSE)
```

Step 9 — Linear Mixed-Effects Models

Because each tree contributes two leaf measurements, a standard linear model would treat those leaves as independent and underestimate standard errors. Linear mixed-effects models (LMEMs) handle this properly by including tree ID as a random intercept, absorbing between-tree variability that is not of primary interest.

The model formula for each response variable y is:

```
y ~ Variety + DCI + LMA + (1|ID)
```

Fixed effects include provenance (Variety), infestation index (DCI), and leaf structural trait (LMA). The random effect (1|ID) allows each tree to have its own baseline level. Models are fitted using lmerTest::lmer() with REML = TRUE. The lmerTest package extends lme4 by providing p-values for fixed effects using Satterthwaite's degrees of freedom approximation. Models are wrapped in tryCatch() so the script continues if any model fails to converge. The response variables modelled are: PI, CCI, Fv/Fm, and A_sat.

Step 10 — Spearman Correlation Matrix Heatmap

A pairwise Spearman correlation matrix is computed for ten variables: Fv/Fm, CCI, DCI, RDB, F_W, D_W, T_W, L_Area, LDMC, and LMA. Only complete cases are used. The matrix is visualised as a heatmap using GGally::ggcorr() with correlation coefficients displayed inside each cell (rounded to 2 decimal places). The colour scale runs from light blue (weak positive) through white (zero) to dark blue (strong positive), consistent with the overall colour theme of the study. Saved as fig6_correlation_matrix.png.

Step 11 — DCI Formula Validation

As a quality-control step, the DCI values stored in the data frame are independently recalculated from their raw components using the Gehring et al. (2018) formula. The maximum absolute deviation between stored and recalculated values is printed — it should be less than 0.000001 if the data generation and formula are consistent. Additionally, a simple linear regression (DCI ~ Sd + Bdor + Gons) is fitted and its R2 is printed; a value of ~1.0 confirms that the three components fully explain the DCI with no residual variance, validating the formula implementation.

```
dci$DCI_check <- (dci$Sd * 0.479 + dci$Bdor * 0.525 + dci$Gons * 0.120) * 100
dci$DCI_diff  <- abs(dci$DCI - dci$DCI_check)
cat(sprintf("Max deviation: %.6f\n", max(dci$DCI_diff, na.rm=TRUE)))
```

3. Results

The following figures summarise the key results of the analysis. All plots were generated using the R script described in Section 2 and exported at 150 dpi.

Figure 1 — Infestation Index (DCI) by Provenance

The violin and boxplot combination shows the distribution of DCI scores across the five provenances. Sicily and Hortiatis (Greek provenance) show wider distributions and higher median DCI values, while Pellice (Italy) and Coruna (Spain) have lower medians. The ANOVA was not significant ($p > 0.05$), indicating that infestation severity did not differ significantly between provenances in this synthetic dataset. The white dot represents the mean for each provenance.

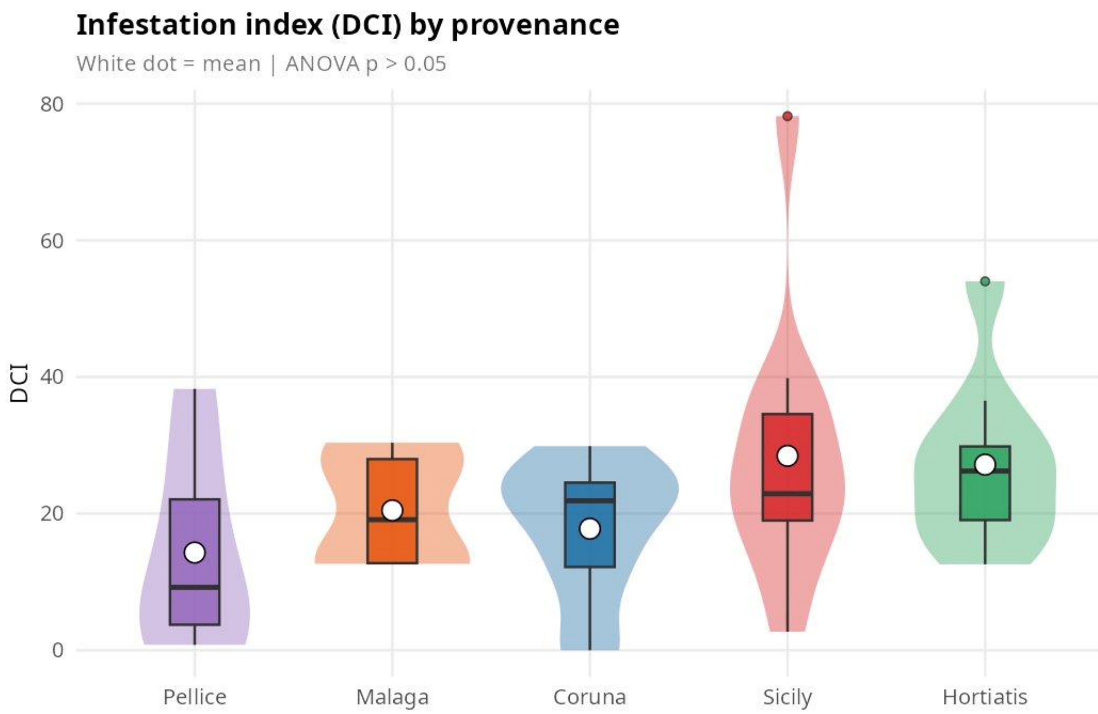

Figure 1. Distribution of the Damage and Crown Index (DCI) by chestnut provenance. Violin outlines show kernel density; inner boxes are standard boxplots; white dots indicate means. ANOVA p > 0.05.

Figure 2 — Reactivated Dormant Buds (RDB) by Provenance

Reactivated dormant buds are a direct plant response to gall wasp attack: when a gall prevents a terminal bud from developing normally, suppressed axillary buds may reactivate. The boxplot shows RDB counts for each provenance. Hortiatis shows the highest median RDB, while Pellice and Sicily have the lowest. The Kruskal-Wallis test was not significant ($p = 0.49$, ns), consistent with the DCI result. Blue dots indicate provenance means.

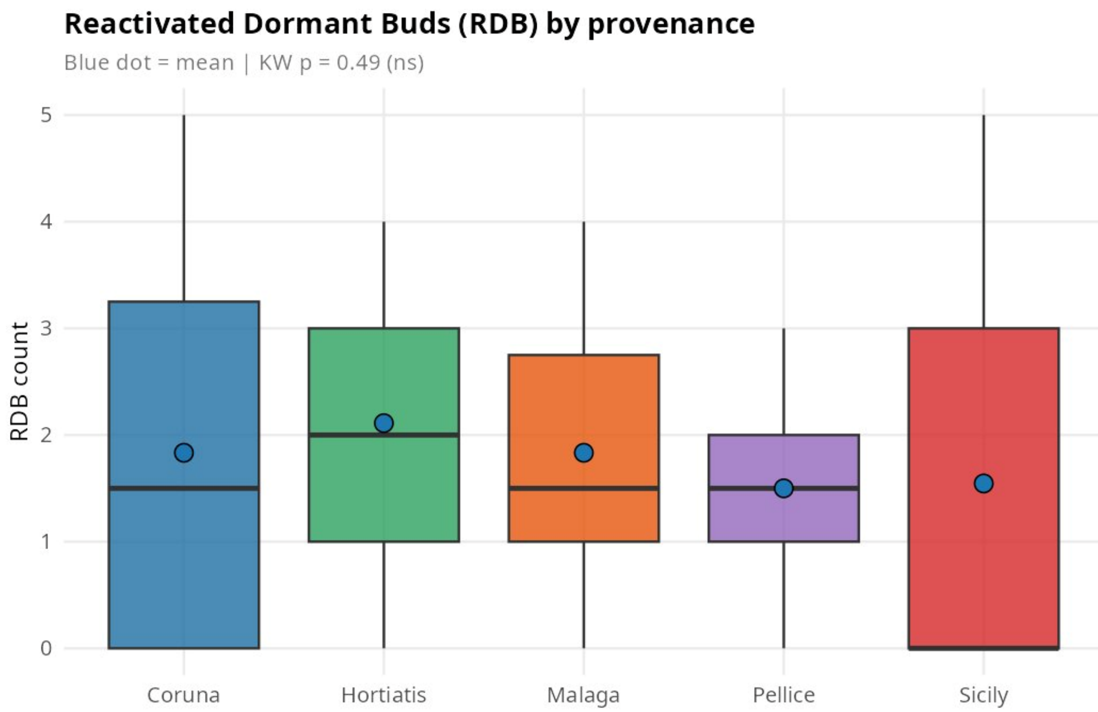

Figure 2. Reactivated dormant bud (RDB) counts by provenance. Blue dots indicate group means. Kruskal-Wallis p = 0.49 (ns).

Figure 3 — Chlorophyll Content Index (CCI)

CCI is a non-destructive proxy for leaf chlorophyll concentration and photosynthetic potential. The left panel shows CCI distributions by provenance (ANOVA p = 0.1720, not significant), and the right panel aggregates provenances by country of origin. Sicily shows a notably lower median CCI and a tight distribution, while Spain (Coruna + Malaga combined) has the broadest range. No significant differences were detected between provenances or countries in this dataset.

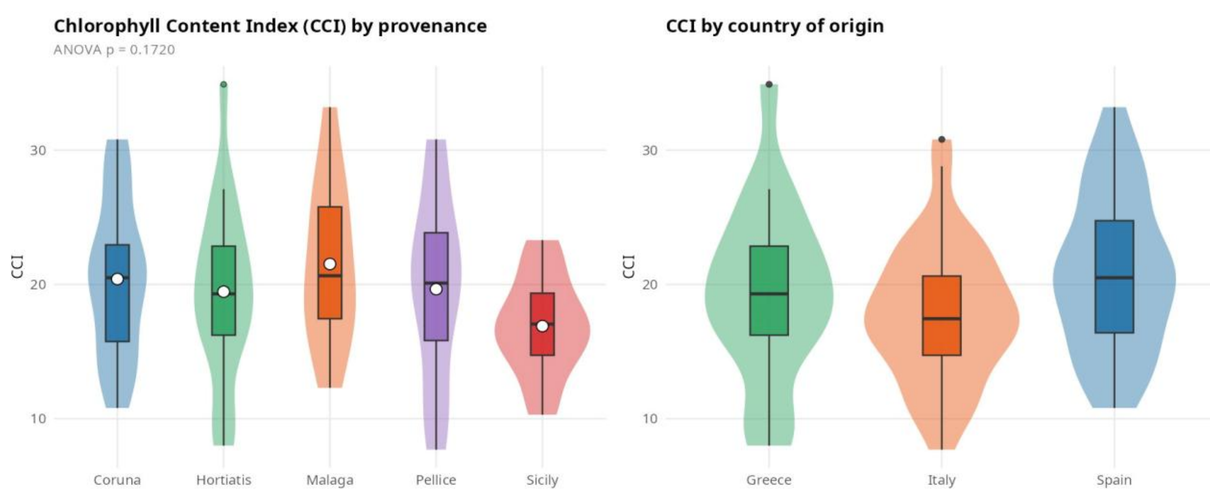

Figure 3. Chlorophyll Content Index (CCI) by provenance (left) and country of origin (right). White/coloured dots indicate means. ANOVA p = 0.1720 (ns).

Figure 4 — PCA Biplot: All Physiological Traits + DCI

The full PCA biplot includes nine variables: PI, Fv/Fm, CCI, Gs, Trm, A_sat, LMA, LDMC, and DCI. Dim1 explains 22.6% and Dim2 explains 19.1% of total variance, meaning roughly 42% is captured in this two-dimensional view. DCI loads negatively on Dim1, opposite to gas exchange variables (Gs, PI), consistent with a negative association between infestation and plant physiological performance. LMA and A_sat load positively on both dimensions. Points are coloured by provenance; note the broad scatter within each group, reflecting high inter-individual variability.

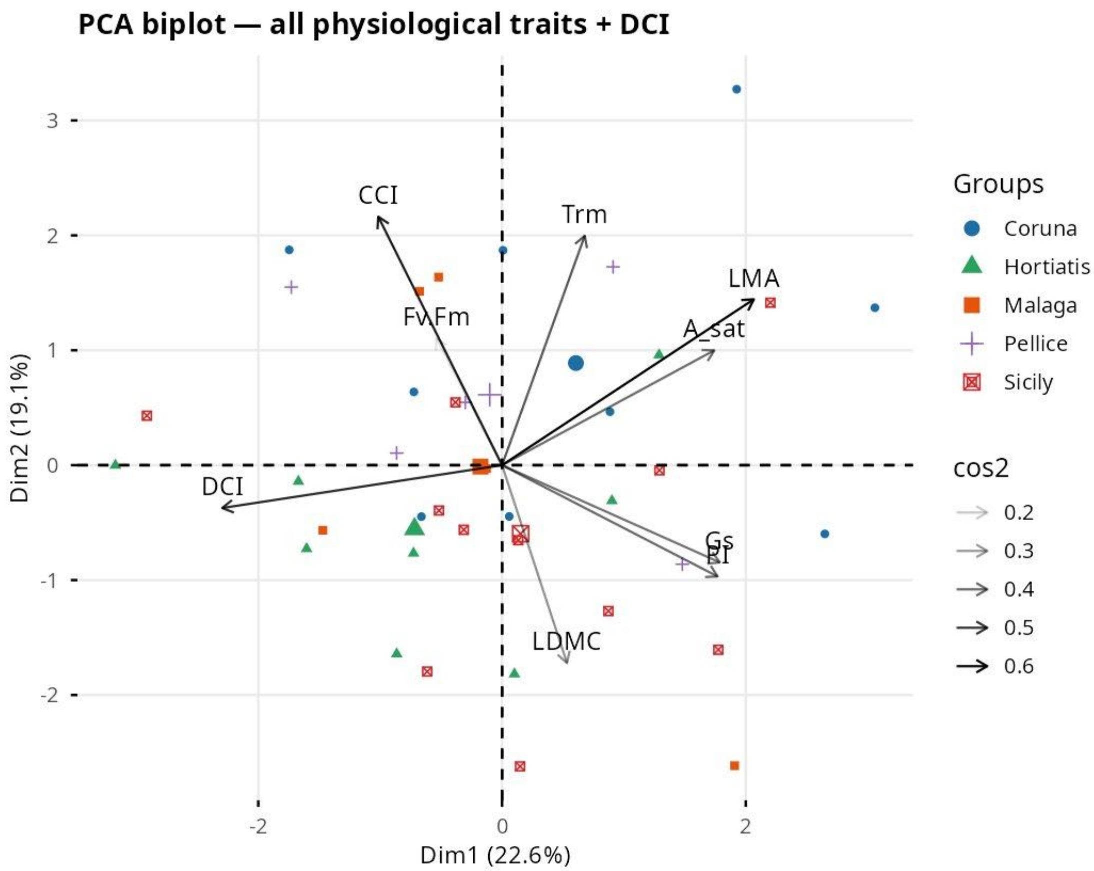

Figure 4. PCA biplot for all physiological traits plus DCI. Each point represents one tree. Arrow opacity scales with cos2 (quality of variable representation on these axes). Dim1 = 22.6%, Dim2 = 19.1%.

Figure 5 — PCA Biplot: Selected Traits

The reduced PCA retains the six most ecologically interpretable variables (PI, CCI, A_sat, LMA, LDMC, DCI). Dim1 now explains 30.2% and Dim2 explains 24.4% of variance — a higher percentage than the full PCA, as removing weakly contributing variables concentrates the explained variance on fewer axes. DCI points in the opposite direction from A_sat, LMA, and CCI, reinforcing the interpretation that higher infestation is associated with lower photosynthetic capacity and leaf structural investment. LDMC loads strongly on the negative Dim2 axis, suggesting a separate structural axis of variation.

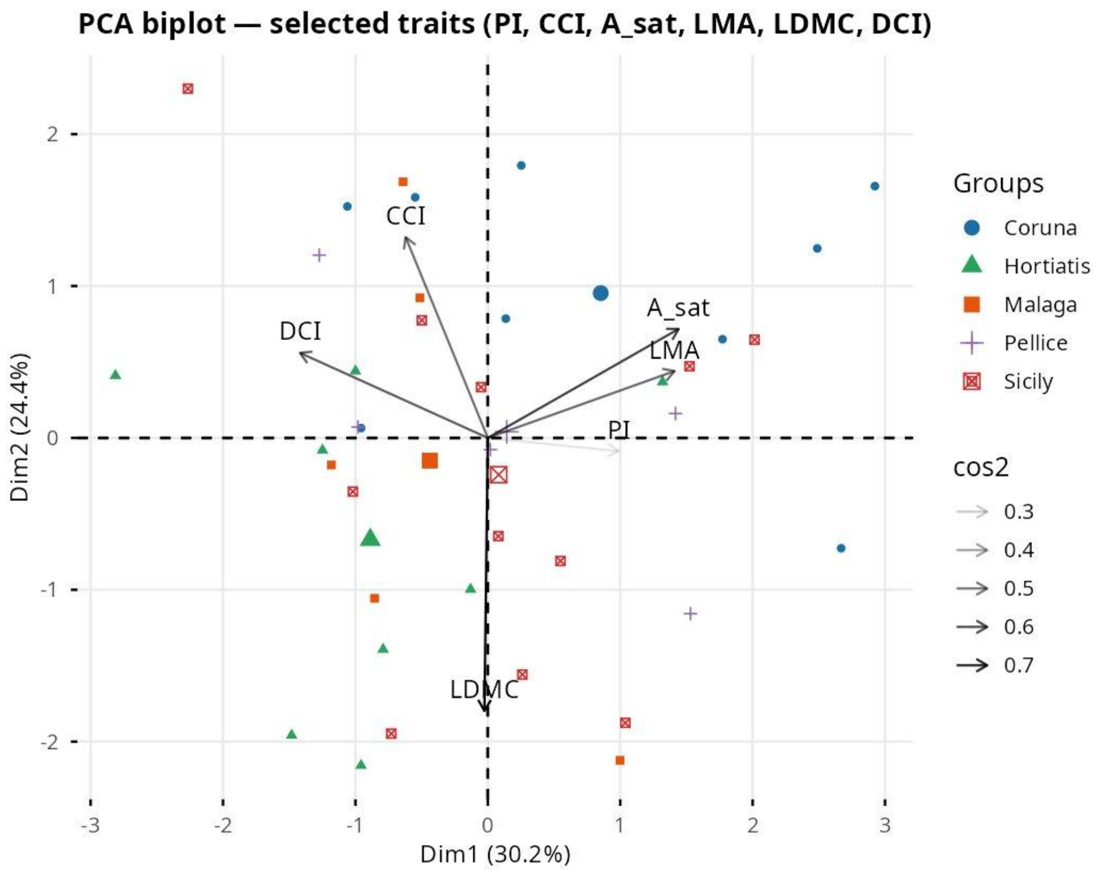

Figure 5. PCA biplot for selected traits (PI, CCI, A_sat, LMA, LDMC, DCI). Dim1 = 30.2%, Dim2 = 24.4%. Arrow opacity scales with cos2.

Figure 6 — Spearman Correlation Matrix

The correlation heatmap reveals the pairwise Spearman relationships among ten variables. The most striking pattern is the very high correlation between F_W, D_W, and T_W (rho > 0.94), which is expected since all three are weight measurements of the same leaf and co-vary tightly. L_Area also correlates positively with the weight variables (rho ~ 0.56-0.65) because larger leaves weigh more. LDMC shows a moderate negative correlation with L_Area (rho = -0.13) and LMA a moderate positive one with D_W (rho = 0.65). DCI shows only weak correlations with leaf physiology variables (|rho| < 0.35), consistent with the non-significant ANOVA results in Steps 4-6.

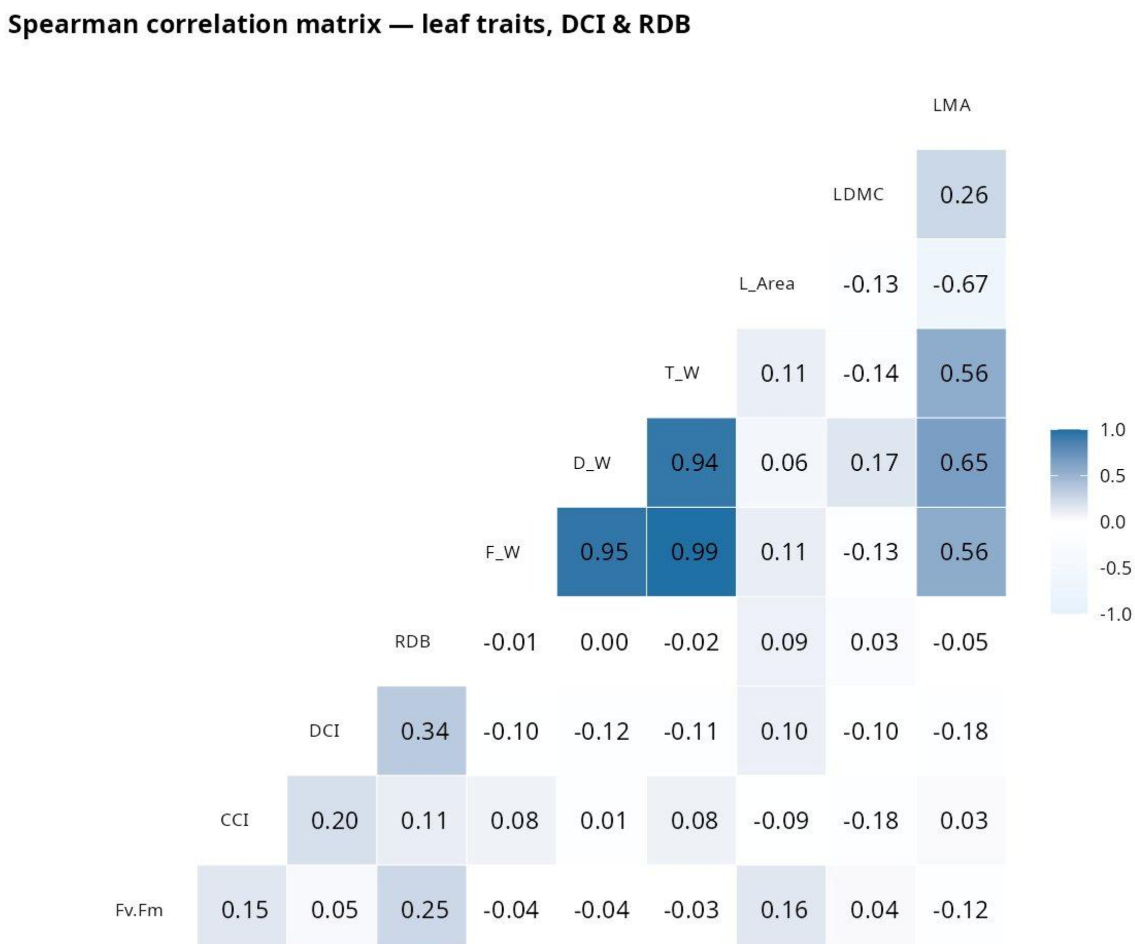

Figure 6. Spearman correlation matrix for leaf traits, DCI, and RDB. Colour intensity and cell labels indicate the strength and direction of correlation. Computed on complete cases only.

References

Gehring, E., Bellosi, B., Quacchia, A., & Conedera, M. (2018). Assessing the Dryocosmus kuriphilus impact on the chestnut tree: the Damage and Crown Index. Journal of Pest Science, Springer.

R packages used: ggplot2, dplyr, tidyr, car, gplots, FactoMineR, factoextra, GGally, lme4, lmerTest, patchwork.

All analyses performed in R (≥ 4.0). Script and synthetic data available at: github.com/KonGianniou